RAGINI SINGHAL

Curriculum Vitae
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Email : raginisinghal1016@gmail.com/ rsinghal@uni-muenster.de
Webpage : raginisinghalmath.github.io

Employment

Oct 2024-2027	Postdoctoral researcher, **University of Münster** Mentor - Hans-Joachim Hein (funded by the Cluster of Excellence for the University of Münster).
Sep 2023-2024	Research associate, **Université Libre de Bruxelles** Mentor - Joel Fine
March-July 2023	Visiting researcher, **Humboldt-Universität zu Berlin** Mentor - Thomas Walpuski
Nov 2021-2022	Research Associate, **King's College London** Mentor - Simon Salamon (funded by the Simons Collaboration on Special Holonomy).

Education

2017-2021	Ph.D. in Mathematics, **University of Waterloo**, Waterloo, Canada. Advisors - Benoit Charbonneau Spiro Karigiannis
2015-2016	M.Sc. in Pure Mathematics, **Imperial College London, UK**. M.Sc. Thesis - Stable minimal cones in Euclidean space. M.Sc. Thesis Advisor - André Arroja Neves
2011-2015	BS in Mathematics, **Indian Institute of Technology, Kanpur**, India. Undergraduate Thesis - Application of knot theory to detect chirality of molecules. Undergraduate Thesis Advisor - Aparna Dar

Research Interests

Riemannian geometry, Geometric analysis, Gauge theory, Deformation theory, Metrics with special holonomy, Lie groups, Spin geometry.

Preprints

(2) S. Dwivedi, R. Singhal, Solutions and singularities of the Ricci-harmonic flow and Ricci-like flows of G_2-structures arXiv

(1) D. Festi, D. Platt, R. Singhal, Y. Tanaka, Examples of real stable bundles on K3 surfaces arXiv

Publications

(4) S. Salamon, R. Singhal, Revisiting 3-Sasakian and G_2-structures, **Ornea, L. (eds) Real and Complex Geometry Springer, Cham.** (2025) arXiv Journal

(3) R. Singhal, *Nearly half-flat structures on* $S^3 \times S^3$, **Differential Geometry and its Applications 97** arXiv:2310.11233, Journal.

(2) R. Singhal, *Deformations of* G_2 *instantons on nearly* G_2 *manifolds*, **Annals of Global Analysis and Geometry** 62, pages 329–366 (2022) arXiv:2101.02151, Journal.

(1) S. Dwivedi and R. Singhal, *Deformation theory of nearly* G_2 *manifolds*, **Communications in Analysis and Geometry** Volume 31 Number 3 2023, arXiv:2007.02497, Journal.

In preparation articles

(6) H.J. Hein, D. Ming, R. Singhal, *Seiberg–Witten theory on toric 4-manifolds.*
(5) S. Salamon, R. Singhal, *SO(4)-invariant nearly parallel* G_2*-structures on Berger Space.*
(4) B. Charbonneau, D. Harland, R. Singhal, *Higher order degormations of instantons on 6-dim Nearly Kähler and Nearly* G_2*-manifolds.*
(3) J. Fine, P. Ghosh, R. Singhal, *Seiberg–Witten equations on Spin(7)-manifolds*
(2) I. Alonso, S. Karigiannis and R. Singhal *Deformation of closed* $SL(n,\mathbb{C})$*-structures with small torsion*
(1) S. Dwivedi and R. Singhal, *Pluripotential theory and cohomology on Spin(7)-manifolds.*

Invited Short-Term Visits

March-June 2024 CRM-Simons Research Visitor, **CRM, Montreal, Canada**

May-June 2014 Summer Research Visitor, **Simon Fraser University, Canada**

May-July 2013 Student Research Fellowship Program 2013, **Indian Statistical Institute, New Delhi, India**

Honours and Awards

- 2023 CRM-Simons Scholar at Centre de Researches Mathématiques, March-June 2024.
- 2021 Doctoral Thesis Completion Award, University of Waterloo, Canada
- 2019 Graduate Studies Research Travel, University of Waterloo, Canada
- 2017-2021 Graduate Research Scholarship, University of Waterloo, Waterloo, Canada
- 2017-2021 International Doctoral Student Award, University of Waterloo, Canada
- 2017-2021 Provost Doctoral Entrance Award for Women, University of Waterloo, Canada
- 2015-2016 Imperial India Foundation Scholarship, Imperial College London,UK
- 2013 Summer Research Fellowship, Indian Academy of Sciences/ Indian National Science Academy/National Academy of Sciences, India
- 2011-2015 Kishor Vaigyanik Protsahan Yojana (KVPY) scholarship, Indian Academy of Sciences, Government of India

Teaching

Instructor & Coordinator

Spring 2020	MATH 117 - Calculus I for Engineers

Teaching Assistant

Winter 2024	KG-2024 Kähler Geometry
Winter 2021	PMATH 450/650 Lebesgue integration and Fourier analysis
	PMATH 365/465 Differential geometry
Fall 2020	PMATH 331 Applied Real Analysis
	MATH 127 Calculus 1 for the Sciences
Winter 2020	PMATH 450/650 Lebesgue integration and Fourier analysis
	MATH 235 Linear Algebra (tutorial center)
Fall 2019	PMATH 365/465 Differential geometry
	MMT 647 Foundations of Calculus I
Spring 2019	PMATH 321 Non Euclidean geometry
	PMATH 351 Real Analysis
Winter 2019	PMATH 333 Introduction to real analysis
	MATH 118 Calculus 2 for Engineering
Fall 2018	MATH 235 Linear Algebra
	PMATH 331 Applied real analysis
Spring 2018	MMT 648 Foundations of Calculus II
Winter 2018	PMATH 365/465 Differential geometry
	MATH 235 Linear Algebra
Fall 2017	PMATH 365/465 Differential geometry
	MATH 147 Calculus I (Advanced level)

Services

- Organiser, Geometry, HOlonomy and Structures with Torsion (GHOST) 2026, 23 – 27 March 2026, Münster, Germany.
- Referee, Journal of Differential Geometry (International press).
- Referee, Quarterly Journal of Mathematics (Oxford University Press).
- Organizer, Workshop: Special Riemannian metrics in dimensions 6,7,8, Centre de Recherches Mathématiques, Montreal, 22-26 April 2024.
- Organizer, British Isles Graduate Workshop, Inverness, 3-7 July 2023.
- Organizer, Junior Special Geometers Meeting, King's College London, 6-8 January 2022.
- Volunteer instructor, Shri Vidhya Jyoti Foundation, Jaipur, India.
- Volunteer, Mathematica Centrum Contest tutor, K-W Bilingual School, Waterloo.
- Judge, Science Fair, K-W Bilingual School, Waterloo.
- Co-organizer, Pure Mathematics Graduate Student Colloquium, University of Waterloo.
- Graduate Volunteer, The Great Polytope Barn-Raising Project, University of Waterloo.
- Grader, CEMC Math contest, University of Waterloo.

Invited Talks

- "Solutions and singularities of the Ricci-harmonic flow", Differential Geometry Seminar, 3 June 2026, University of Waterloo.
- "Cohomogeneity-one nearly G_2-structures" , Forschungsseminar über Differentialgeometrie, 6 January 2025, University of Hamburg, Germany.
- "Nearly half-flat structures on $S^3 \times S^3$", Mathematisches Seminar, 17 December 2024, University of Kiel, Germany.
- "Deformation theory of nearly G_2-instantons" Geometric Models of Matter 2024, 21 August 2024, Leeds University, UK.
- "Cohomogeneity-one nearly G_2-structures" Special geometries in Dimension 6,7,8, 26th April 2024, CRM Montreal
- "Deformation theory of G_2-structures and instantons", Young Scholar day, 20 December 2023, Belgian Math Society.
- "Nearly half-flat structures on $S^3 \times S^3$", Oberseminar, November 27, 2023, Universität Münster.
- "Deformation theory of nearly G_2 manifolds", ULB Differential Geometry Seminar, May 2, 2023, Université Libre de Bruxelles.
- "Deformations of G_2 instantons on nearly G_2 manifolds", Oxford-London Gauge assembly, November 11, 2022 (online).
- "Deformations of G_2 instantons on nearly G_2 manifolds", Leeds geometry seminar, September 21 2022, University of Leeds.
- "Deformations of G_2 instantons on nearly G_2 manifolds", UCL geometry seminar, February 16 2022, UCL.
- "Deformations of G_2 instantons on nearly G_2 manifolds", Simons collaboration workshop "Connections between String Theory and Special Holonomy", January 10-14 2022, Oxford.
- "Deformation theory of nearly G_2 manifolds", Differential Geometry Seminar 2020-2021, UC Santa Barbara.
- "Deformation of instantons on nearly G_2 manifolds", AMS Fall Eastern Virtual Sectional meeting, 2020.
- "Deformation of instantons on nearly G_2 manifolds", Ottawa Mathematics Conference 2020, Ottawa.
- "Deformation theory of nearly G_2 manifolds", KCL/UCL Junior Geometry Seminar, 2020.
- "Deformation of instantons on nearly G_2 manifolds" **IIT Kanpur, India**; 11/11/2019
- "Deformations of G_2 instantons on nearly G_2 manifolds", G_2 geometry and related topics, **CMO-BIRS, Oaxaca, Mexico**; 07/05/2019
- "Minimal and Willmore surfaces", Graduate Student Colloquium, **University of Waterloo**, Canada; 26/03/2019

References

- Hans-Joachim Hein, University of Münster, (hhein@uni-muenster.de)
- Joel Fine, Université Libre de Bruxelles, (joel.fine@ulb.be)
- Simon Salamon, King's College London, (simon.salamon@kcl.ac.uk)
- Derek Harland, University of Leeds, (d.g.harland@leeds.ac.uk)
- Benoit Charbonneau, University of Waterloo (bcharbon@uwaterloo.ca)
- Spiro Karigiannis, University of Waterloo (karigiannis@uwaterloo.ca)